

TM Forum Technical Specification

Bill Calculation Management

TMFC031

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Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

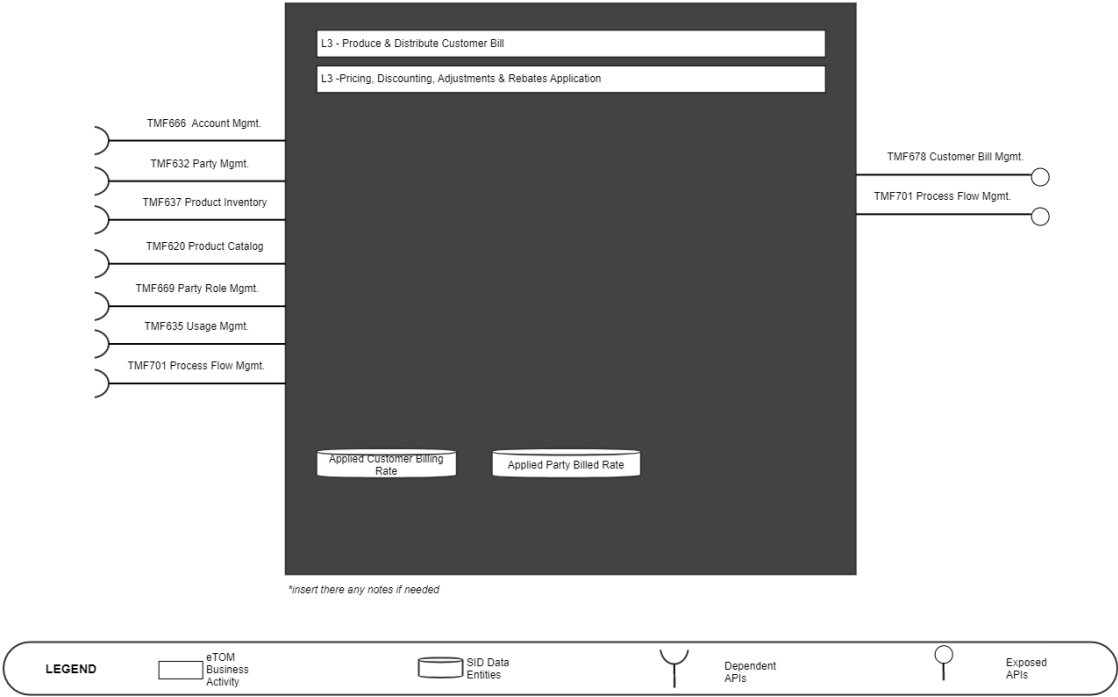
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1. Overview

Component Name	ID	Description	ODA Function Block
Bill Calculation	TMFC031	The Bill Calculation processes all charges against billing accounts during bill cycles. Bill Calculation can be executed both on a cyclic basis and on demand. It performs calculations with bill compilation of charges, credits, fees & taxes, including pro rata, at various levels, such as product and/or account level that have been generated since the last run for that account, applying promotions and discounts as well.	Core Commerce Management

TMFC031: Bill Calculation



2. eTOM Processes and SID Data Entities

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for.

Identifier	Level	Business Activity Name	Description
1.3.9	L2	Customer Bill Invoice Management	Ensure the bill invoice is created, physically and/or electronically produced and distributed to customers, and that the appropriate taxes, discounts, adjustments, rebates and credits for the products and services delivered to customers have been applied.
1.3.9.4	L3	Pricing, Discounting, Adjustments & Rebates Application	Ensure that the bill invoice is reflective of all the commercially agreed billable events and any bill invoice adjustments agreed between a Service Provider and the customer.
1.3.9.4.1	L4	Obtain Billing Events	Accept billing events that have been collected, translated, correlated, assembled, guided and service rated before determining the information would be applied to the customer's bill invoice(s).
1.3.9.4.2	L4	Apply Pricing, Discounting, Adjustments & Rebates to Customer Account	Determine the customer account or customer specific pricing, charges, discounts, and taxation that should be delivered to the invoice(s) for the customer.
1.3.9.4.3	L4	Apply Agreed Customer Bill Adjustment	Apply and review any adjustment agreed in the previous billing period and make these included to the bill invoice.
1.6.15	L2	BP Bill/Invoice Management	Business Partner Bill/Invoice Management manages the Business Partner bill/invoice process, controls bills/invoices, manages the lifecycle of bills/invoices. A bill is a notice for payment which is supposed to be preceded by an invoice in most cases.
1.6.15.1	L3	BP Bill/Invoice Process Management	Make certain that there is capability so that the Bill Invoice Management processes can operate effectively and design and develop an enterprise's invoicing process.
1.6.15.3	L3	BP Bill/Invoice Lifecycle Management	Ensure bills/invoices are created, physically and/or electronically produced and distributed to parties, and that the appropriate taxes, discounts, adjustments,

Identifier	Level	Business Activity Name	Description
			rebates and credits for the products delivered to parties have been applied.
1.6.15.3.1	L4	Apply BP Pricing, Discounting, Adjustments & Rebates	Ensure that a bill/invoices is reflective of all the commercially agreed billable events and any bill/invoice adjustments agreed between an enterprise and a BP.
1.6.15.3.5	L4	Receive BP Bill/Invoice	Receive and record the bill/invoice from a BP . Compare a BP bill/invoice against all transactions with the BP that would result in a bill/invoice being sent to the enterprise. Manage the interactions between a BP and an enterprise. Approve a BP bill/invoice

2.2. SID ABEs

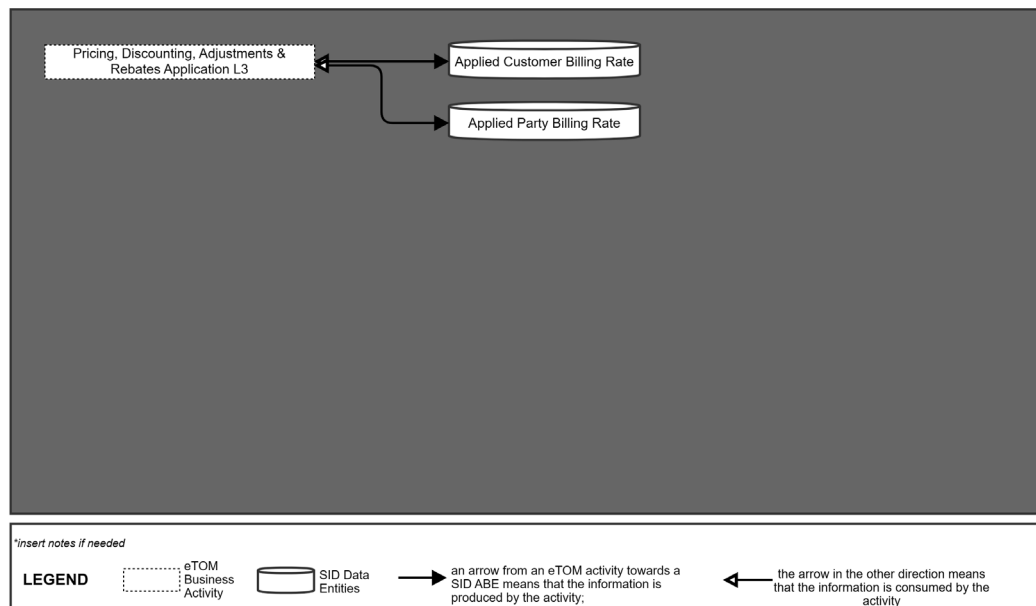
SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)*
Applied Customer Billing Rate ABE	n/a
Applied Party Billing Rate ABE	n/a

*: if SID ABE Level 2 is not specified this means that all the L2 business entities must be implemented, else the L2 SID ABE Level is specified.

2.3. eTOM L2 - SID ABEs links

eTOM L2 vS SID ABEs links for this ODA Component.



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
67	Usage Summary and Details Presentation	Usage Summary and Details Presentation presents usage summary and details (billed and non-billed) for a specific time period.	Invoice Management	Invoicing
87	Billing Event Processing Guiding	Billing Event Processing Guiding support for a consistent processing.	Invoice Management	Invoicing
316	Billing Administration	The Billing Administration function manages the data that are necessary to perform the bill calculation: billing cycle data, management of runs, groups and cycles of invoicing.	Invoice Management	Invoicing
399	Billing Management Integration	Billing Management Integration provide a Virtual Network Operators online access function to make them self-sufficient for Billing management.	Invoice Management	Invoicing

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		Including the use of VNO/Dealer data fencing.		
32	Billing Initialization	Billing Initialization initializes the bill and sends to the Bill Calculation application the required information for accounts that are going to be processed.	Invoice Management	Invoicing
72	Billing Account Price Plan Determining	Billing Account Price Plan Determining associates a charge record with the appropriate price plan.	Invoice Management	Billing Account Administration
70	Charge To Billing Account Distribution	Charge To Billing Account Distribution identifies the related prepaid or postpaid billing account for a given charge (recurring, one time, usage).	Invoice Management	Billing Account Administration
69	Charge To Billing Account Identification	Charge To Billing Account Identification associates incurred charge to the billing account liable for its payment.	Invoice Management	Billing Account Administration
68	Charges to Billing Statement Identification	Charges to Billing Statement Identification identifies what charges are to be included in the statement.	Invoice Management	Billing Account Administration
158	Commitment Tracking Result Determining	Commitment Tracking Result Determining determines the outcome of the evaluation (financial benefits or penalties) in the context of the gathered data for commitment tracking.	Rating and Follow up	Bill Calculation
159	Commitment Tracking Terms & Conditions Evaluation	Commitment Tracking Terms & Conditions Evaluation evaluates the terms and conditions in the context of the gathered data for commitment tracking.	Rating and Follow up	Bill Calculation

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
160	Commitment Tracking Data Collection	Commitment Tracking Data Collection collects data to be used in the evaluation of the terms and conditions to monitor financial commitments between the customer and the provider.	Rating and Follow up	Bill Calculation
256	Customer Bill Usage and Charges Viewing	Customer Bill Usage and Charges Viewing provides an internet technology driven interface to the customer to undertake Usage and charges comparison and unbilled charges view directly for themselves.	Rating and Follow up	Bill Calculation
89	Billing Event Aggregation	Billing Event Aggregation is part of the Billing Event Processing to supply aggregated billing events to the Billing System.	Invoice Management	Invoicing
90	Billing Event Processing Analyzing	Billing Event Processing Analyzing provides billing event analysis and billing event aggregations analysis to control the usage data sent to the Billing System.	Invoice Management	Invoicing
183	Bill Charges Aggregation	Bill Charges Aggregation function determines charges (including recurring, one time and usage charges) for purchased products and services in a given bill run based on the customer price plan set at time of the customer order/contract negotiation.	Rating and Follow up	Bill Calculation
60	Split Bill Charge Distribution	Split Bill Charge Distribution provides charge and event distribution to support a split bill.	Rating and Follow up	Tariff Calculation and Rating
184	Currency Conversion	Currency Conversion identifies the required currency conversion if any	Rating and Follow up	Bill Calculation

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2
		needed to appropriately bill the customer.		
61	On Demand Bill Calculation	On Demand Bill Calculation function will invoke a bill calculation on demand for e.g. a purchase.	Rating and Follow up	Bill Calculation
300	Discounts Calculation*	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.	Rating and Follow up	Tariff Calculation and Rating
55	Price and Discount Calculation*	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).	Rating and Follow up	Tariff Calculation and Rating

**Discounting may apply to different levels and for this component it refers to end of cycle discounts that cannot be managed or applied in Product Configurator or Product Usage Management.*

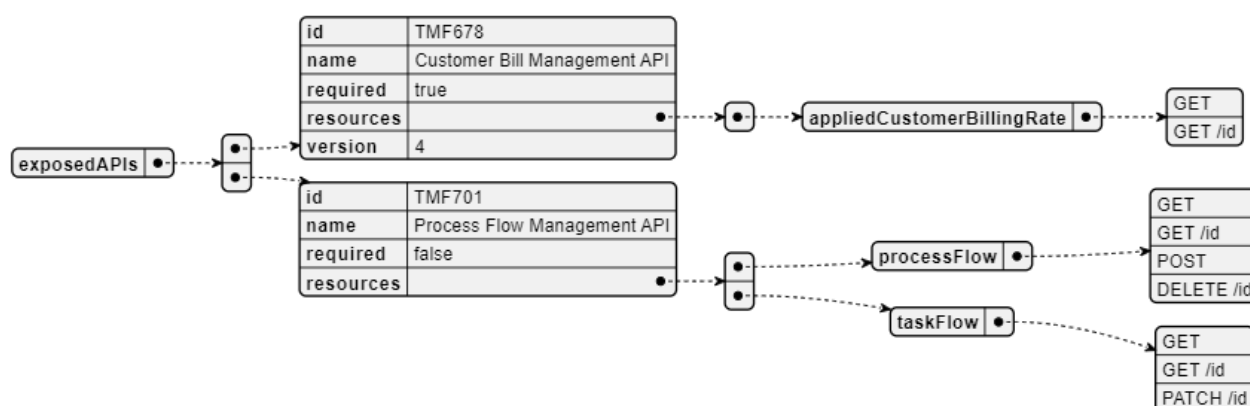
3. TMF OPEN APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate is listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

Following diagram illustrates API/Resource/Operation:



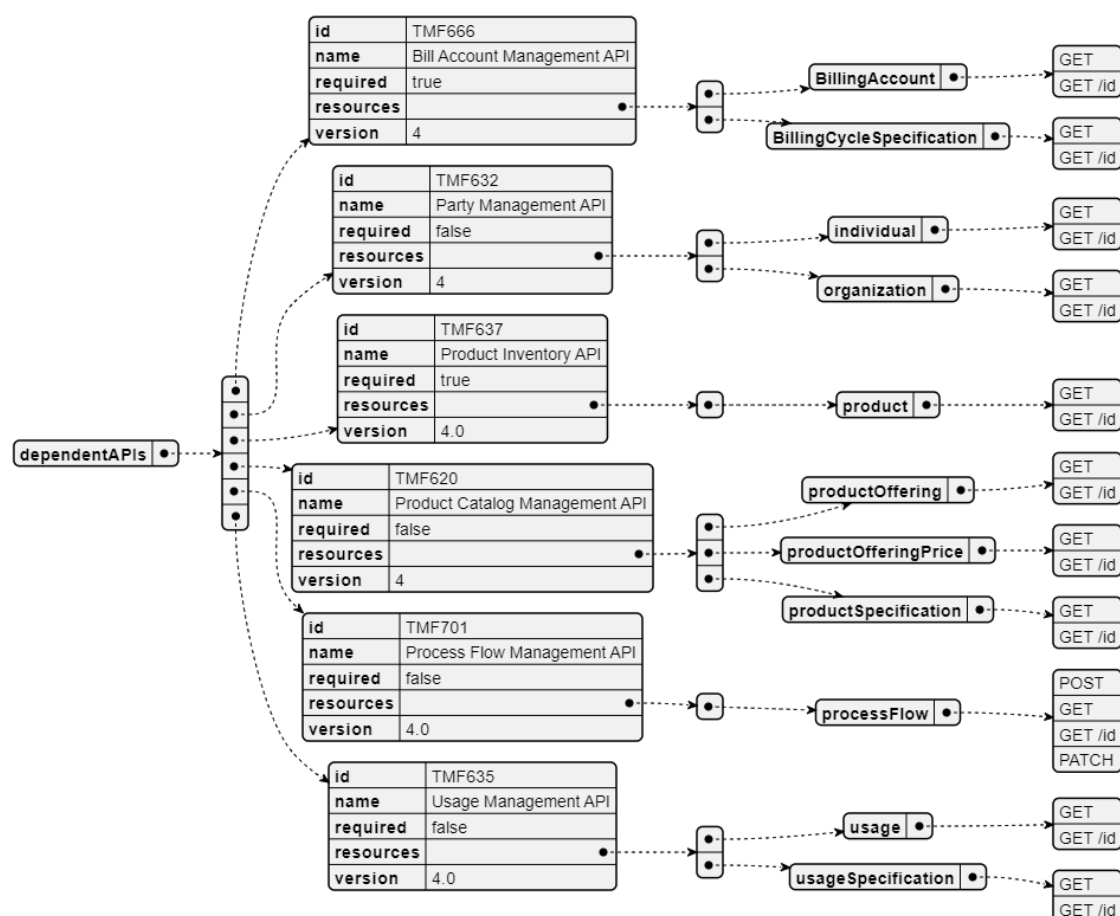
API ID	API Name	API Version	Mandatory / Optional	Operations
TMF678	Customer Bill Management	4	Mandatory	appliedCustomerBillingRate: • GET • GET /id
TMF701	Process Flow Management	4	Optional	processFlow: • GET • GET /id • POST • DELETE /id taskFlow: • GET

API ID	API Name	API Version	Mandatory / Optional	Operations
				- GET /id - PATCH /id

Note: TMF678 only for applied customer bill resource

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation potentially used by the product catalog component:

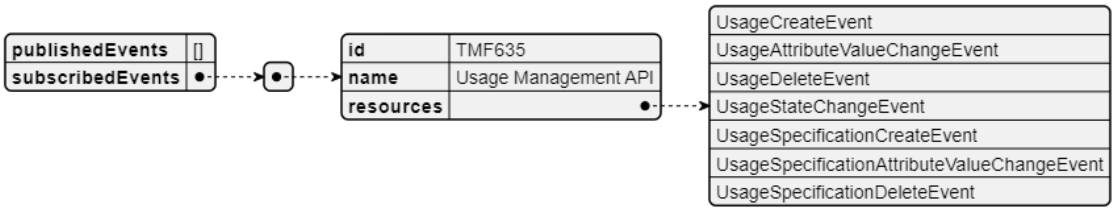


API ID	API Name	API Version	Mandatory / Optional	Operations	Rationales
TMF666	AccountManagement	4	Mandatory	billingAccount: GET, GET /id, POST, PATCH /id, DELETE /id billingCycleSpecification: GET, GET /id, POST, PATCH	Billing Account and Billing Cycle information required to understand what BillingAccount to apply the

API ID	API Name	API Version	Mandatory / Optional	Operations	Rationales
				/id, DELETE /id billingFormat: GET, GET /id, POST, PATCH /id, DELETE /id billing PresentationMedia: GET, GET /id, POST, PATCH /id, DELETE /id	customer rate to and the Billing Cycle period.
TMF6 32	PartyManagement	4	Optional	individual: GET, GET /id organization: GET, GET /id	
TMF6 37	ProductInventory	4	Mandatory	product: GET, GET /id	To retrieve installed product and price information to determine the applied rate.
TMF6 20	Product Catalog	4	Optional	productOffering: GET, GET /id productOfferingPrice: GET, GET /id productSpecification: GET, GET /id	
TMF6 69	PartyRoleManagement	4	Optional	partyRole: GET, GET /id	
TMF6 35	Usage Management	4	Optional	usage: GET, GET /id usageSpecification: GET	
TMF7 01	ProcessFlowManagement	4	Optional	processFlow: GET, GET /id, POST, PATCH /id	

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
SID	23.5
eTOM	23.5
Functional Framework	23.5

5.2. Jira References

5.3. Further resources

- a. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	27 Sep 2023	Gaetano Biancardi	First version of document
1.0.0	19 Dec 2023	Amaia White	Final administrative edits
1.1.0	29 Jan 2024	Elisabeth Andersson	Functional Framework: • 6 Unbilled Invoice Items Listing > added • 68 Charges to Billing Statement Identification > added • 69 Charge To Billing Account Identification > added • 70 Charge To Billing Account Distribution > added • 158 Commitment Tracking Result Determining > added • 159 Commitment Tracking Terms & Conditions Evaluation > added • 160 Commitment Tracking Data Collection > added • 256 Customer Bill Usage and Charges Viewing > added
1.1.0	01 Mar 2024	Amaia White	Final administrative edits prior to release
2.0.0	29 May 2024	Elisabeth Andersson Gaetano Biancardi	• Updated to the latest template • Added TMF620 Product Catalog as dependent API • Modified the optionality of APIs: ◦ From Optional → Mandatory: TMF637

Version Number	Date Modified	Modified by:	Description of changes
			○ From Mandatory → Optional: TMF669, TMF635 ○ Removed TMF672 and TMF688 as they will be part of Canvas Services ○ Removed TMF669 ○ Removed the need to access billingFormat and billingPresentationmedia from TMF666 ○ Updated TMF666 billingAccount and billingCycleSpecification to only have Get and Get/ID • Updated to 23.5 versions of frameworks • Updates on eTOM: ○ Removed 1.3.9.3 ○ Added 1.6.15.3.1 ○ For clarification added higher level descriptions for already included lower levels for 1.6.15 • Updated on FF: ○ Change to Aggregate function levels for 158, 159, 160, 256 ○ Added: 60, 61, 89, 90, 183, 184
2.0.0	06 Sep 2024	Amaia White	Final administrative edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	19 Dec 2023	Amaia White	Initial release
Pre-production	22 Jan 2024	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	01 Mar 2024	Amaia White	Release of v1.1.0

Release Status	Date Modified	Modified by:	Description of changes
Production	26-Apr-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	06 Sep 2024	Amaia White	Release of v2.0.0
Production	01-Nov-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgments

Team Member (@mention)	Company	Role*
Gaetano Biancardi	Accenture	Author
Sylvie Demarest	Orange	Reviewer
Elisabeth Andersson	Matrixx	Key Contributor

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer